

# **CHF, Gender, and Hemodynamics: Modeling Post-MI Mortality**

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# About the Dataset

- 500 samples
- 14 health features
- Death
- Right-censored

|   | afb | age  | av3 | bmi   | chf | cvd | diasbp | gender | hr   | los  | miord | mitype | sho | sysbp |
|---|-----|------|-----|-------|-----|-----|--------|--------|------|------|-------|--------|-----|-------|
| 0 | 1   | 83.0 | 0   | 25.54 | 0   | 1   | 78.0   | 0      | 89.0 | 5.0  | 1     | 0      | 0   | 152.0 |
| 1 | 0   | 49.0 | 0   | 24.02 | 0   | 1   | 60.0   | 0      | 84.0 | 5.0  | 0     | 1      | 0   | 120.0 |
| 2 | 0   | 70.0 | 0   | 22.14 | 0   | 0   | 88.0   | 1      | 83.0 | 5.0  | 0     | 1      | 0   | 147.0 |
| 3 | 0   | 70.0 | 0   | 26.63 | 1   | 1   | 76.0   | 0      | 65.0 | 10.0 | 0     | 1      | 0   | 123.0 |
| 4 | 0   | 70.0 | 0   | 24.41 | 0   | 1   | 85.0   | 0      | 63.0 | 6.0  | 0     | 1      | 0   | 135.0 |

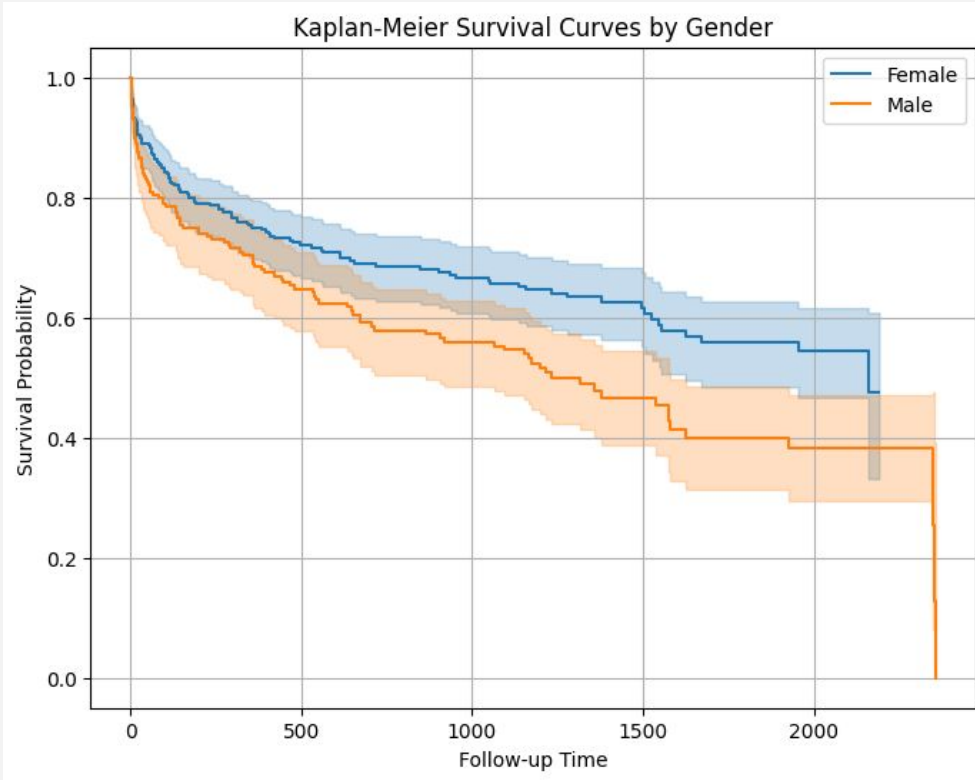
# Explanation of Variables

- **age**: Age at Hospital Admission, a key risk factor for cardiovascular events.
- **bmi**: **Body Mass Index** ( $\text{kg/m}^2$ ), used to classify obesity and its prognostic role in heart failure.
- **sysbp**: **Initial Systolic Blood Pressure** (mmHg), the pressure in arteries during heartbeats.
- **diasbp**: **Initial Diastolic Blood Pressure** (mmHg), the pressure in arteries between heartbeats.
- **hr**: **Initial Heart Rate** (beats per minute).
- **gender**: Patient's sex (female/male, coded as 0/1).
- **los**: **Length of Hospital Stay** in hospital (days), an outcome measure.
- **afb**: **Atrial Fibrillation** (AF), an irregular heart rhythm increasing stroke risk.
- **chf**: **Congestive Heart Complications**, structural heart disease with symptoms like fluid retention.
- **miord**: **Myocardial Infarction Order**, indicating order of **heart attack** events (e.g., first vs. recurrent MI).
- **mitype**: **Myocardial Infarction Type** (e.g., STEMI, NSTEMI, or Q-wave/non-Q-wave MI classification).
- **sho**: **Cardiogenic Shock**, a critical condition post-MI or in heart failure.
- **cvd**: **History of Cardiovascular Disease** (categorical).
- **av3**: **Complete Heart Block** (categorical).
- **lenfol**: **Total Length of Follow-up** (time from admission until either death or censor).
- **Fstat**: **Vital Status at Last Follow-up** (event of death).

# Log-Rank Test(Stratified)

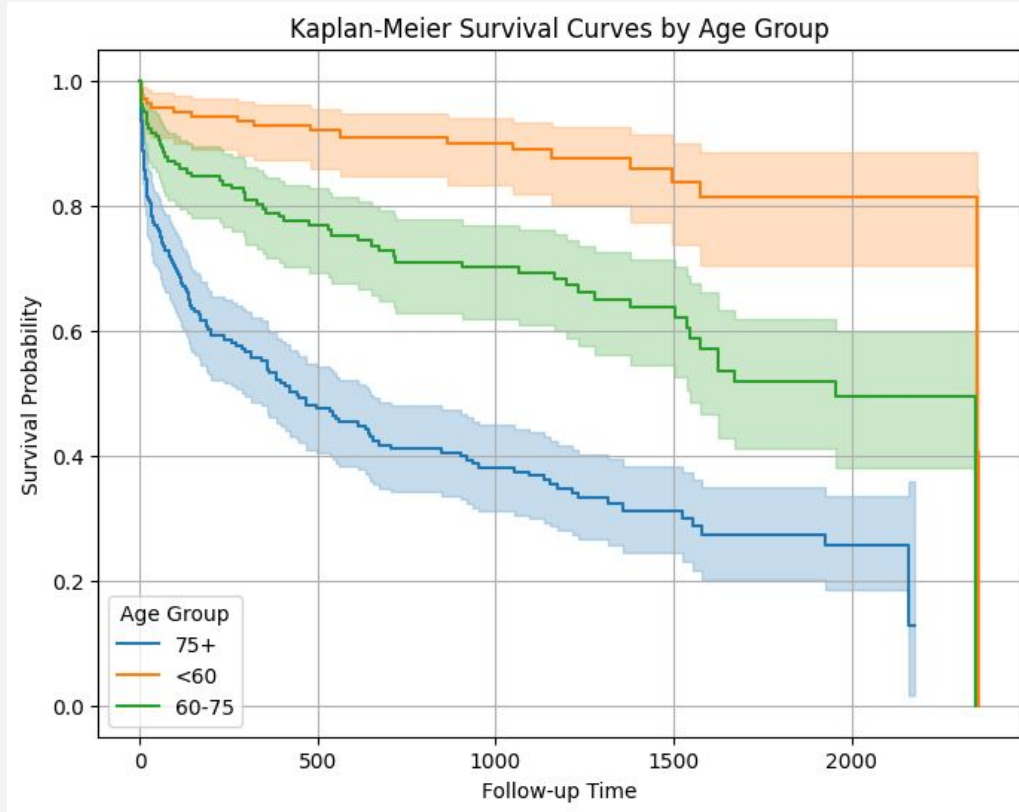
| Variable | df  | Chi-sq   | p-value | Significance    |
|----------|-----|----------|---------|-----------------|
| afb      | 2   | 10.9     | 0.001   | significant     |
| age      | 66  | 111.2762 | 0       | significant     |
| av3      | 2   | 1.5236   | 0.2171  | not significant |
| bmi      | 411 | 26.4456  | 0       | significant     |
| chf      | 2   | 84.6007  | 0       | significant     |
| cvd      | 2   | 2.8615   | 0.0907  | not significant |
| diasbp   | 97  | 22.5229  | 0       | significant     |
| gender   | 2   | 7.7911   | 0.0053  | significant     |
| hr       | 105 | 29.1571  | 0       | significant     |
| los      | 27  | 0.6668   | 0.4142  | not significant |
| miord    | 2   | 9.5706   | 0.002   | significant     |
| mitype   | 2   | 16.1806  | 0.0001  | significant     |
| sho      | 2   | 23.8365  | 0       | significant     |
| sysbp    | 133 | 7.538    | 0.006   | significant     |

# Kaplan-Meier by Gender



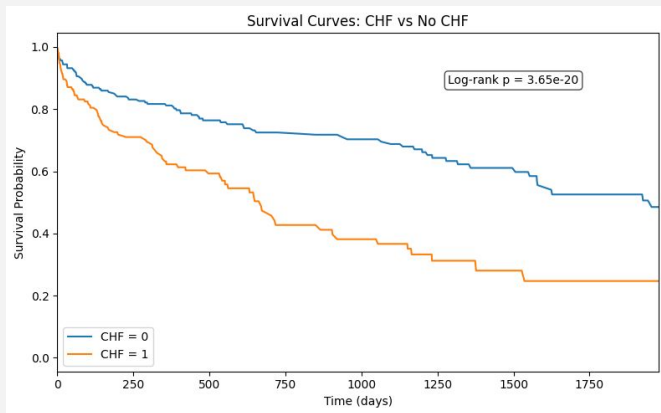
- Kaplan–Meier curves show a **difference in survival between males and females**
- One group exhibits **consistently lower survival probabilities over time**
- The separation of curves begins **early and persists throughout follow-up**
- Log-rank test indicates a **statistically significant difference** ( $p = 0.005$ )

# Kaplan-Meier by Age Group



- Survival probability **decreases as age increases**
- Patients aged **75+ have the lowest survival**, while those under 60 have the highest
- Clear and consistent **separation between age groups over time**
- Indicates a strong relationship between **age and mortality risk**
- Log-rank test shows **highly significant differences across groups** ( $p < 0.001$ )

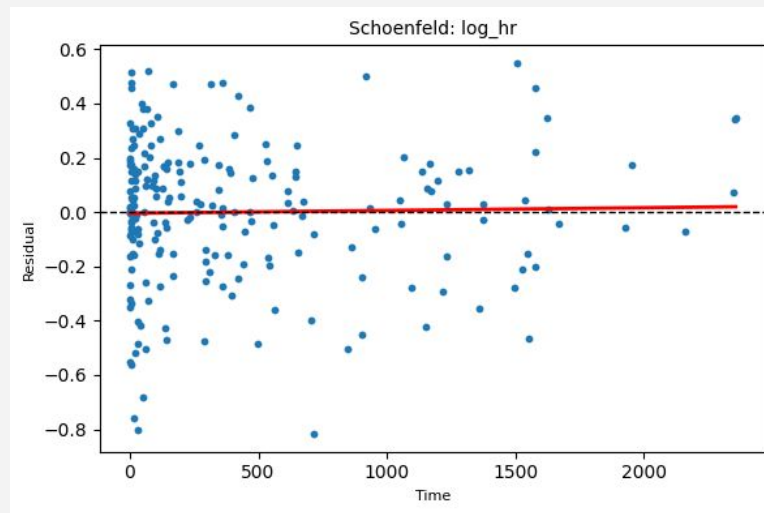
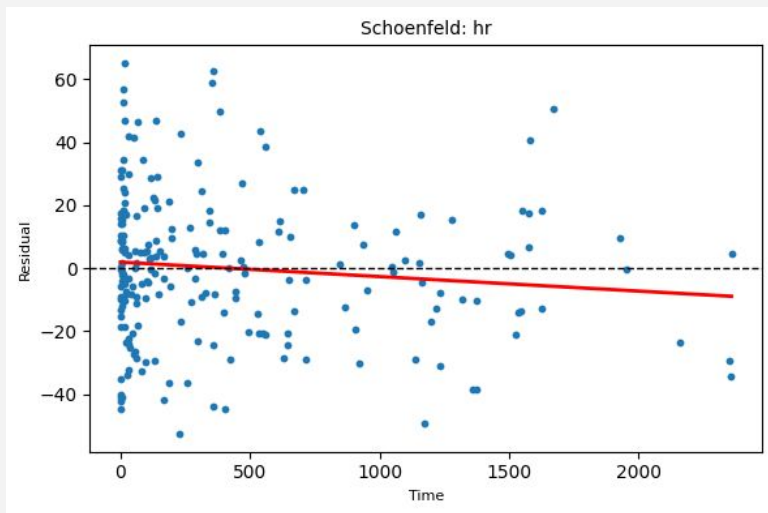
# Cox Model & Congestive Heart Failure Survival



| Covariate               | Hazard Ratio | p-value |
|-------------------------|--------------|---------|
| Gender                  | 0.699        | 0.016   |
| Age                     | 1.048        | <0.001  |
| HR (per 1 SD)           | 1.355        | <0.001  |
| BMI (per 1 SD)          | 0.759        | 0.001   |
| Diastolic BP (per 1 SD) | 0.782        | 0.002   |

- Dropped av3 (11/500 cases, unstable)
- Removed los: post-attack hospital time is a response variable, not a predictor; circular
- Univariate log-rank → Insignificant Hazards: afb, cvd, sysbp, miord
- PH violations → stratified sho, mitype
- CHF patients show dramatically lower survival - log-rank  $p = 3.65e^{-20}$

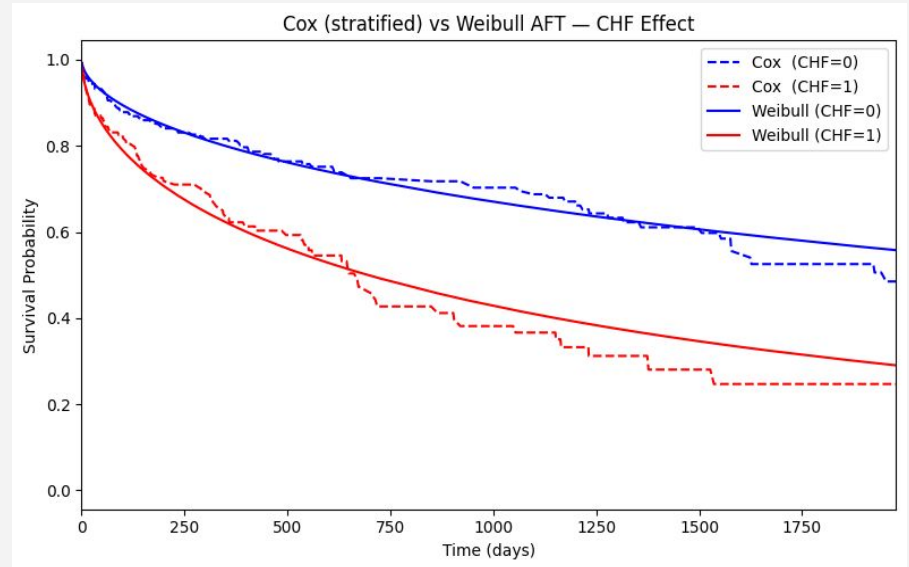
# Schoenfeld Diagnostics



- HR and BMI showed declining residual trends → log-transformed → trends resolved
- CHF residuals split into two bands → effect is structural, not scalar → stratified
- All remaining covariates flat after corrections → PH satisfied, no extended Cox needed

# Weibull Accelerated Failure Time : Robustness Check

- CHF stratified out of Cox → no direct Hazard Rate estimable from Cox alone
- Weibull AFT has no PH assumption → CHF enters directly as a covariate
- Both models agree: CHF effect is large, significant, and consistent
- CHF patients experience time 3.9x faster - expected survival reduced to ~26% of non-CHF baseline



# Key Insights

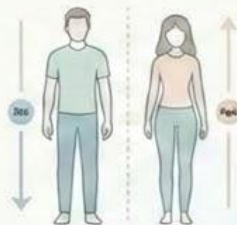
- Both **gender and age significantly influence survival outcomes**
- **Age emerges as a stronger predictor** of survival compared to gender
- Survival patterns show **early divergence**, indicating immediate risk differences
- Findings are **consistent with clinical expectations** for heart attack patients
- Results provide insight into **high-risk patient groups**
- An increase in **Heart Rate** strongly links to a reduction in survivability
- **Diastolic BP** and **BMI** (obesity paradox) increases correlate to a reduced hazard
- **Congestive Heart Failure** leads to an increase in hazard, and is robust across methodologies

# Survival Analysis Project: Key Insights

Summary of findings on post-MI mortality factors.



## Gender & Age: Significant Influencers

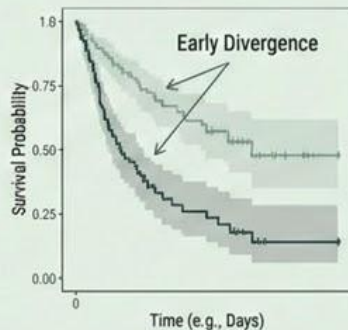


Both gender and age significantly influence survival outcomes.

Age emerges as a stronger predictor of survival compared to gender.

## Survival Patterns & Clinical Expectations

- Survival patterns show early divergence, indicating immediate risk differences.
- Findings are consistent with clinical expectations for heart attack patients.



## Physiological Risk Factors



Increased Heart Rate links to reduced survivability.



Higher Diastolic BP and BMI correlate to a reduced hazard (obesity paradox).



Congestive Heart Failure leads to a robust increase in hazard.